## HW 2

Please refer to the exercises in the textbook: Quarteroni, Sacco and Saleri, Numerical Mathematics, 2nd edition. **Please submit your code, numerical results, and provide necessary explanation of your observations for Q2 and Q3.**

1. Recall the cubic spline interpolation mentioned in class.

   (a) Consider instead using a quadratic spline. Write down the form of this spline and try to impose continuity conditions for the interpolation and derivatives. How many (continuous) derivatives can it have? How many extra conditions need to be added?

   (b) Generalize to splines using polynomials of degree k.

2. (Newton form)

   (a) Write a routine `newt_polyval(nodes, coeffs, xvals)` that evaluates the interpolant in Newton form,

   $$p(x) = \sum_{i=0}^{n-1} c_i \prod_{j=0}^{i-1} (x - x_j).$$

   The coefficients as input shall be of the form `coeffs =` $(c_0, \cdots, c_{n-1})$.

   (b) Use the provided divided difference code (page 342 of textbook) together with the evaluate function in (a) to check if it agrees with the 'polyfit' function in Matlab.

3. (Chebyshev nodes)

   (a) Use your Newton form code in the previous question or Matlab 'polyfit' function to test

   $$f(x) = \frac{1}{1 + ax^2} \quad \text{in } [-1, 1]$$

   with equally spaced nodes. Verify divergence when $a = 25$ but convergence when $a = 1$ and show a log-plot of the error.

   (b) Consider the function

   $$f(x) = \frac{1}{1 + 25x^2} \quad \text{in } [-1, 1]$$

   Test what happens for the equally spaced interpolation nodes and Chebyshev nodes

   $$x_i = \cos\left(\frac{\pi(2i+1)}{2n}\right), \quad \text{for } i = 0, \ldots, n-1.$$

   respectively.

   (c) For the function in (b), use cubic spline interpolation for the equally spaced interpolation node, and compare the cubic spline results with the interpolation results from (b).